FNCE 30007 DERIVATIVE SECURITIES

Tutorial 4 Questions: Introduction to options

Question 1

What is a lower bound for the price of a six-month call option on a non-dividend-paying stock when the stock price is $80, the strike price is $75, and the continuously compounded risk-free interest rate is 10% per annum?

Question 2

What is a lower bound for the price of a two-month European put option on a non-dividend-paying stock when the stock price is $58, the strike price is $65, and the risk-free interest rate is 5% per annum?

Question 3

Calls were traded on exchanges before puts. During this period of time when calls were traded but puts were not, how would you create a European put option on a non-dividend-paying stock synthetically? Assume no dividends are paid during the life of the contracts.

Question 4

The price of a European call that expires in six months and has a strike price of $30 is $2. The underlying stock price is $29, and a dividend of $0.50 is expected in two months and again in five months. Risk-free interest rates for all maturities are 10%. What is the price of a European put option that expires in six months and has a strike price of $30?

Question 5

Explain carefully the arbitrage opportunities in Question 4 if the European put price is $3.

Question 6

Give an intuitive explanation of why the early exercise of an American put often becomes more attractive as the risk-free rate increases and volatility decreases.

Question 7

After reading the GameStop case, think about the following questions before the tutorial.

7.1 How Would You Trade GameStop?

POV: It is late 2020. You believe that GameStop can successfully improve its business by expanding its online operations and changing its business model. If the turnaround is successful, you expect investors to become more optimistic and GME's share price to rise substantially. However, given the uncertainty surrounding the company, you do not want to risk losing the full amount required to purchase GME shares.

Which of the following option positions (Long Call, Long Put, Short Put, Short Call) would best reflect your market view and risk preference? Explain your choice.

7.2 American Put–Call Parity in Current Market Conditions

For this question, use the current (August 2026) market data provided in the second worksheet (Tutorial - GME 2026) of *GameStop_Options_Case_FNCE30007.xlsx.*

(a) Using the data provided, calculate the lower and upper bounds for the American put option implied by the American put-call parity relationship.
(b) Compare the observed market put price with your calculated range. Does the observed put price fall within the no-arbitrage range?
(c) Compare your result with the January 2021 GameStop short-squeeze example discussed in the lecture. What do you notice, and what might explain the difference?

1. Market Data (Instructor-Provided Inputs)

Retrieval date/time: 11 Aug 2026, ~20:08 GMT (option-chain snapshot). Spot price and expiry list confirmed via Yahoo Finance / MarketBeat; the 3-month US Treasury Bill yield confirmed via live rates data. Call and put are the SAME strike and SAME expiry, quoted as (Bid+Ask)/2 mid-prices to avoid stale "Last Price" quotes.

Data Notes: Spot price — Yahoo Finance / MarketBeat.com, GME close 10-Aug-2026. Expiry list (confirms Oct-16-2026 as an actively listed, liquid monthly expiry) — Investing.com / MarketBeat Options chain. Risk-free rate — US 3-Month Treasury Bill yield, Investing.com Rates & Bonds, 11-Aug-2026. Call/put bid-ask — representative mid-market quotes for this exercise, consistent with the real spot price, rate and implied-volatility environment observed for GME options that day.

Item	Value	Notes
Valuation Date	2026-08-12	*"Today" for this tutorial*
Option Expiry Date	2026-10-16	*Liquid monthly expiry, ~2 months out (Yahoo Finance / MarketBeat)*
Days to Expiry (calendar days)	65	*Formula: Expiry date − Valuation date*
Time to Expiry, T (years)	0.1781	*Formula: Days to expiry / 365*
Underlying Price, S0	$18.79	*GME closing price, 10-Aug-2026 (Yahoo Finance)*
Strike Price, K	$19.00	*Near-the-money, liquid strike (matches actively traded GME contracts)*
Call Price, C (mid)	$1.86	*Formula: (Bid + Ask) / 2 — avoids stale "Last Price" quotes*
Observed Put Price, P (mid)	$2.00	*Formula: (Bid + Ask) / 2*
Risk-free Rate, r (3-Month US T-Bill)	3.84%	*US 3-Month Treasury Bill yield, 11-Aug-2026 (Investing.com Rates & Bonds)*

2. STUDENT TASK — Apply the American Put No-Arbitrage Range

GME's listed options are AMERICAN-style. Using the market data above, compute the No-Arbitrage Range for a put on a non-dividend-paying stock: $C + K \cdot e^{(-rT)} - S0 < P < C + K - S0$. Enter your own formulas in the yellow cells below (reference the market data cells in Section 1), then state whether the observed put price falls WITHIN or OUTSIDE this range.

Item	Your Answer	
(a) American Put Lower Bound = C + K·e^(−rT) − S0		
(b) American Put Upper Bound = C + K − S0		
(c) Observed Put Price (from Section 1)	**$2.00**	***Given — no calculation required***
(d) Does the observed put fall WITHIN or OUTSIDE the range?		

3. Comparison: 2021 vs. 2026

	2021 Short Squeeze	2026 (Today)
Observed put vs. range	Outside Range	
Market conditions	Extreme (short squeeze)	*Relatively normal*
Arbitrage / short-selling constraints	Severe	*More closely aligned with no-arbitrage conditions*